

JIANYE XU

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Education

RWTH Aachen University, Germany PhD, Chair of Embedded Software, Department of Computer Science	2023-04 – now
RWTH Aachen University, Germany MSc, Automation Engineering, Institute of Automatic Control	2020-10 – 2022-09 <i>GPA: 1.2/1.0 MSc with Distinction</i>
RWTH Aachen University, Germany BSc (Exchange Student), Mechanical Engineering, Faculty of Mechanical Engineering	2019-10 – 2020-08
Beijing Institute of Technology, China BSc, Vehicle Engineering, School of Mechanical Engineering	2016-10 – 2019-09 <i>GPA: 91.48/100 Outstanding Graduate (top 5%)</i>

Publications

- J. Beerwerth, **J. Xu**, S. Schäfer, F. Belderink, and B. Alrifaaee. Zero-shot marl benchmark in the cyber-physical mobility lab. *at - Automatisierungstechnik*, 74(5):376–385, 2026. doi: 10.1515/auto-2025-0057.
- A. Mokhtarian, **J. Xu**, P. Scheffe, M. Kloock, S. Schäfer, H. Bang, V.-A. Le, S. Ulhas, J. Betz, S. Wilson, S. Berman, L. Paull, A. Prorok, and B. Alrifaaee. A survey on small-scale testbeds for connected and automated vehicles and robot swarms: A guide for creating a new testbed. *IEEE Robotics & Automation Magazine*, pages 2–19, 2024. doi: 10.1109/MRA.2024.3505772.
- S. Schäfer, **J. Xu**, D. Klüner, A. Mokhtarian, P. Scheffel, and B. Alrifaaee. Educational applications of the cyber-physical mobility lab: A summary. In *2024 European Control Conference (ECC)*, pages 2666–2671, 2024. doi: 10.23919/ECC64448.2024.10591176.
- P. Scheffe, **J. Xu**, and B. Alrifaaee. Limiting computation levels in prioritized trajectory planning with safety guarantees. In *2024 European Control Conference (ECC)*, pages 297–304, 2024. doi: 10.23919/ECC64448.2024.10591179.
- J. Xu** and B. Alrifaaee. High-order control barrier functions: Insights and a truncated taylor-based formulation. *arXiv preprint arXiv:2503.15014*, 2025a. doi: 10.48550/arXiv.2503.15014.
- J. Xu** and B. Alrifaaee. A learning-based control barrier function for car-like robots: Toward less conservative collision avoidance. In *2025 European Control Conference (ECC)*, pages 988–995, 2025b. doi: 10.23919/ECC65951.2025.11187043.
- J. Xu** and B. Alrifaaee. TTCBF: A truncated Taylor control barrier function for high-order safety constraints. *arXiv preprint arXiv:2601.15196*, 2026. doi: 10.48550/arXiv.2601.15196.
- J. Xu**, P. Hu, and B. Alrifaaee. SigmaRL: A sample-efficient and generalizable multi-agent reinforcement learning framework for motion planning. In *2024 IEEE 27th International Conference on Intelligent Transportation Systems (ITSC)*, pages 768–775, 2024a. doi: 10.1109/ITSC58415.2024.10919918.
- J. Xu**, O. Sobhy, and B. Alrifaaee. XP-MARL: Auxiliary prioritization in multi-agent reinforcement learning to address non-stationarity. *arXiv preprint arXiv:2409.11852*, 2024b. doi: 10.48550/arXiv.2409.11852.
- J. Xu**, C. Che, and B. Alrifaaee. A real-time control barrier function-based safety filter for motion planning with arbitrary road boundary constraints. In *2025 IEEE 28th International Conference on Intelligent Transportation Systems (ITSC)*, pages 2818–2825, 2025. doi: 10.1109/ITSC60802.2025.11423203.
- J. Xu**, J. Betz, A. Mokhtarian, A. Mittal, M. Cai, R. Mangharam, O. M. Shehata, C. M. Elias, J.-N. Zaech, P. Scheffe, F. Jahncke, S. S. Ulhas, K. M. Arfvidsson, and B. Alrifaaee. Small-scale testbeds for connected and automated vehicles and robot swarms: Challenges and a roadmap. *IEEE Intelligent Transportation Systems Magazine*, 2026.

Awards

Best Poster Award 2025 German Autonomy Summit (GAS)	2025-12
Best Paper Nomination IEEE International Conference on Intelligent Transportation Systems (ITSC)	2025-09
Dean's List (top 5% in the academic year 2021/2022) RWTH Aachen University	2023-01
MSc with distinction (GPA 1.2/1.0) RWTH Aachen University	2022-09
German National Scholarship RWTH Aachen University	2021-11

Patent: Quick Locking and Unlocking Device for Vehicle-Mounted Power Battery Box (first inventor, publication number: CN 111086379 B) China National Intellectual Property Administration **2021-06**

German National Scholarship RWTH Aachen University **2020-11**

Outstanding Graduate (GPA 91.48/100, top 5%) Beijing Institute of Technology **2020-06**

China National Scholarship The Ministry of Education of the People's Republic of China **2018-11**

Second Prize in National Student Mechanical Product Digital Design Competition China Mechanical Discipline Steering Committee **2018-06**

Outstanding Undergraduate Beijing Institute of Technology **2017-12**

First Prize in Beijing Student Engineering Design Expression Competition Beijing Municipal Commission of Education **2017-12**

First Prize in National Student Drafting and Modelling Innovation Competition China Cartographic Association **2017-07**

Experiences

Assistant Lecturer *RWTH Aachen University, Germany* **Winter terms 2024 & 2025**

- Master's Course: "Control and Perception in Networked and Autonomous Vehicles"
- <https://cpm.lrt.unibw.de/courses/master/>

Workshop Invited Speaker *IEEE ITSC, Gold Coast, Australia* **2025-11**

- Workshop: "Automated Mobility in Mixed Traffic—Emerging Mixed Autonomy in the Era of AI"
- <https://www.itsc2025.mixedtraffic.org/>

Workshop Co-Organizer *IEEE ITSC, Gold Coast, Australia* **2025-11**

- Workshop: "Automated Mobility in Mixed Traffic—Emerging Mixed Autonomy in the Era of AI"
- <https://www.itsc2025.mixedtraffic.org/>

Guest Lecture *RWTH Aachen University, Germany* **2025-01**

- Guest Lecture: "Control Barrier Functions and Their Applications to Collision Avoidance in Networked and Autonomous Vehicles"

Workshop Co-Organizer *IEEE Intelligent Vehicles Symposium (IV) 2024, Jeju, South Korea* **2024-06**

- Workshop: "1st Workshop on Small-Scale Testbeds for Connected and Automated Vehicles and Robot Swarms"
- <https://cpm.lrt.unibw.de/workshop/>

Competition Co-Organizer *CPM Olympics, Aachen, Germany* **2024-01 – 2024-06**

- Competition: A motion planning competition for connected and automated vehicles
- <https://cpm.lrt.unibw.de/olympics/>

Beijing Institute of Technology *Undergraduate Teaching Assistant* **2017-09 – 2017-10**

- Task: delivered CAD software training lectures to first-year students.